We use the NYC Building Energy and Water Disclosure dataset, published by the New York City government. It contains building characteristics and reported annual greenhouse-gas emissions.

Each row represents a property's report for one calendar year. A property may contain one building or several buildings. The same property can appear in different years.

Our downloaded CSV contains 103,259 records for 2022–2024 and 16 selected attributes:

Attribute	Description
report_year	Calendar year of the report
property_id	Unique identification number of the property
parent_property_id	Identification number of its parent property or campus
primary_property_type_self	Main purpose of the property, such as office or hotel
largest_property_use_type_1	Floor area of the largest use, in square feet
_2nd_largest_property_use_1	Floor area of the second-largest use, in square feet
_3rd_largest_property_use_1	Floor area of the third-largest use, in square feet
year_built	Year the building was constructed
number_of_buildings	Number of buildings included in the property
occupancy	Percentage of floor area occupied
borough	Administrative area within New York City
total_location_based_ghg	Reported annual greenhouse-gas emissions, in metric tonnes CO_2 equivalent
estimated_values_energy	Indicates whether estimated energy values were used
alert_energy_meter_has_less	Flags meter data covering less than 12 full calendar months
alert_energy_meter_has_gaps	Flags missing periods in meter records
metered_areas_energy	Indicates which areas are covered by energy meters

Why this dataset is useful: It connects property characteristics—such as size, purpose, and occupancy—with reported emissions across many properties.

What We Do with This Dataset

We use this dataset to predict the expected annual greenhouse-gas emissions of a property based on its characteristics.

We provide eight inputs to the model: property type, three floor-area values, construction year, number of buildings, occupancy, and borough.

The model learns from the reported emissions of existing properties. We use CatBoost Regression because the output is a numerical value.

Example: For an office property with particular floor areas, occupancy, age, and location, the model might predict 500 tonnes CO_2e per year.

This prediction becomes a peer-emission benchmark—an estimate based on properties with similar characteristics. We can then compare it with the property's reported emissions.

It is an expected emission value, not a government-approved emission limit.

We clean the dataset by checking missing values, duplicate properties, estimated energy readings, and incomplete meter coverage, retaining suitable records from 2024. We fill missing numerical inputs using training-data medians and transform the emission target using log(1 + emissions) before training.

We divide the properties into four separate sets: 60% for training, 15% for validation, 10% for calibration, and 15% for testing. These sets are used to learn emission patterns, choose model settings, estimate prediction uncertainty, and evaluate performance on unseen properties, respectively.

We train CatBoost Regression using the eight input attributes to predict annual greenhouse-gas emissions, and use validation data to choose the model settings. We then retrain the selected model on the combined training and validation data.

We use the separate calibration data to calculate a prediction interval around each estimated emission value, showing its uncertainty. For example, a prediction of 500 tonnes CO_2e may have an interval of 350–700 tonnes CO_2e; we later check how often these intervals contain the reported test emissions.

We evaluate the model on unseen test properties using MAE, RMSE, and R^2, and check the prediction intervals' coverage and width. We compare CatBoost with Random Forest and a simple median baseline to assess whether it improves emission predictions.